

GANGSHAN, Taiwan

WMO: 467450

Lat: 22.783N

Lon: 120.262E

Elev: 34

StdP: 14.68

Time Zone: 8.00 (E08)

Period: 86-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	49.9	52.2	42.8	40.6	52.2	45.9	45.8	54.9	22.9	64.0	20.3	65.5	7.1	0	0.352

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	91.8	80.9	91.2	80.8	89.9	80.4	82.7	88.6	82.2	88.1	81.6	87.5	11.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.0	161.1	86.3	80.6	159.2	86.1	80.0	155.9	85.6	46.9	89.4	46.2	88.3	45.4	87.8	88.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.4	16.4	14.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			43.0	85.7	3.1	1.8	40.8	87.0	39.0	88.1	37.3	89.1	35.1	90.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	75.8	65.1	66.3	70.7	76.1	80.4	82.9	83.9	83.3	81.9	78.1	72.9	66.9
	DBStd	7.70	4.19	4.91	5.38	4.53	2.75	2.48	2.05	1.98	2.20	2.89	4.05	4.25
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	143	51	40	17	3	0	0	0	0	0	0	4	28
	CDD50	9403	468	457	641	783	942	987	1050	1032	958	872	688	525
	CDD65	4071	54	77	193	336	477	537	585	567	508	407	242	88
	CDH74	39316	202	365	1167	2589	4687	6228	7094	6652	5387	3313	1267	365
CDH80	13588	11	36	227	730	1583	2430	2959	2590	1871	922	200	29	
Wind	WSAvg	6.4	8.2	8.0	7.3	5.9	5.6	5.8	6.2	6.1	5.4	5.5	6.6	6.9
Precipitation	PrecAvg	69.8	0.8	0.9	1.0	3.0	6.4	14.7	13.4	16.9	8.9	2.1	1.1	0.7
	PrecMax	111.7	6.4	3.4	4.1	10.0	23.1	42.3	33.0	46.0	42.5	9.5	8.0	5.4
	PrecMin	40.9	0.0	0.0	0.0	0.0	0.0	0.6	0.1	3.7	0.6	0.0	0.0	0.0
	PrecStd	19.5	1.2	1.0	1.2	2.8	4.7	10.6	7.9	11.5	9.0	2.7	1.6	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.9	83.7	86.1	89.9	91.5	93.3	93.6	93.0	91.7	88.3	86.0	82.8
		MCWB	72.3	73.9	76.3	77.5	79.6	80.4	81.4	81.7	80.7	78.4	76.1	73.6
	2%	DB	78.6	80.8	84.3	87.9	89.7	91.5	91.8	91.3	89.9	87.5	84.1	80.4
		MCWB	69.1	71.8	75.1	77.2	79.5	80.5	81.2	81.3	80.2	78.3	75.5	71.3
	5%	DB	76.2	78.6	82.7	86.1	88.2	89.9	91.1	89.9	89.1	86.1	82.4	77.8
		MCWB	67.4	69.7	73.8	76.9	79.2	80.4	81.0	80.9	80.0	77.6	74.1	69.1
	10%	DB	73.7	75.6	80.6	84.3	87.3	89.2	89.7	89.2	87.8	84.6	80.6	75.6
		MCWB	65.4	68.0	72.6	76.0	78.8	80.3	80.6	80.7	79.8	76.4	73.0	67.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.1	74.8	77.7	80.0	82.1	83.5	83.7	83.5	82.8	80.1	78.1	74.4
		MCDB	79.8	81.3	84.5	86.5	87.5	89.0	89.4	89.3	88.2	86.3	84.8	81.5
	2%	WB	70.0	72.8	76.0	78.6	80.8	82.2	82.7	82.6	81.7	79.2	76.2	71.9
		MCDB	76.4	79.3	82.6	85.2	87.3	87.8	88.9	88.5	87.4	85.5	82.6	78.7
	5%	WB	68.2	70.7	74.6	77.6	79.9	81.5	82.0	81.9	80.8	78.2	75.0	70.2
		MCDB	74.4	77.0	81.1	84.3	86.3	87.3	88.3	87.8	86.8	84.4	81.1	76.3
	10%	WB	66.5	68.8	73.3	76.5	79.1	80.8	81.3	81.2	80.0	77.3	73.5	68.6
		MCDB	72.8	75.2	79.6	83.0	85.1	86.8	87.6	86.9	85.7	83.4	79.2	74.3
Mean Daily Temperature Range	5% DB	MDBR	17.2	17.0	16.4	14.2	12.8	11.3	11.5	11.1	11.7	13.6	15.2	17.6
		MCDBR	19.0	19.0	17.7	15.8	14.1	12.8	13.4	12.6	12.8	13.9	15.2	18.5
		MCWBR	11.3	11.4	10.4	8.6	7.4	6.3	6.7	6.4	6.7	7.5	8.5	10.7
	5% WB	MCDBR	17.4	17.8	16.6	14.4	13.1	11.6	12.0	11.7	11.8	13.0	14.2	17.1
		MCWBR	11.1	11.4	10.4	8.5	7.5	6.6	6.7	6.6	6.7	7.6	8.7	10.3
Clear-Sky Solar Irradiance	taub	0.507	0.548	0.663	0.650	0.536	0.494	0.497	0.552	0.591	0.601	0.522	0.526	
	taud	2.013	1.931	1.695	1.768	2.101	2.248	2.233	2.050	1.912	1.841	2.037	1.974	
	Ebn at Noon	237	236	216	222	247	256	255	242	230	219	231	225	
	Edh at Noon	50	58	77	73	52	44	45	54	61	63	49	51	
All-Sky Solar Radiation	RadAvg	1075	1248	1375	1515	1620	1701	1810	1613	1572	1443	1158	1028	
	RadStd	114	139	126	156	180	185	148	178	137	89	91	105	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (21 neighbors)	+0.43	N/A	N/A	N/A	+0.36	+0.43	N/A	N/A	+209	+198

Nomenclature: See separate page